The Magic Thermometer

Student sheet

Name: _____ **Date:** _____

Safety: Do not touch a hot radiator. The box is not waterproof: keep water away from it.

1. What I see on the screen

On the screen you can see the cards **Temperature** (in degrees Celsius) and **Humidity** (in percent), and below them their chart. You start with **Start** and **Let's start!**; you move on with the green button at the bottom right, you go back with **Back**, and the last green button reads **Finish**.

The sensor is slow. After you move the box, you press **I moved the box**, **wait 60–90 seconds**, until the app's timer turns green, and only then write the number down. The box stays on the table, not in your hand, and nobody breathes on it.

2. What I do on my own

First I predict. I do not change my prediction after I measure.

1. Is the whole room the same temperature? ☐ yes, the same everywhere ☐ no, it is different from place to place

I think the table is about _____ °C. Put an arrow on the line, where you think the room is today:

0 °C	20 °C	37 °C	100 °C
freezing	comfortable room	my body	water boils

2. Heat source (circle one): computer radiator lamp warm water. Beside it I think it will be:

☐ warmer ☐ cooler ☐ the same by about _____ °C

because _____

_____

While I wait, I draw the map of the room. One dot for every place I will measure, with its name beside it. After I measure, I write the temperature next to each dot: blue if it is cool, red if it is warm.

Now I measure. I put the box down, wait **60–90 seconds**, press **Capture** twice, 30 seconds apart, then write the numbers on the screen. If I do not write decimals, I write only the whole number. The last two rows are the **vertical map**: same corner of the room, different height.

Place	Temperature (°C)	Humidity (%)	Circle one
On the table (the room)			cool / warm / hot
Beside the heat source			cool / warm / hot
On the floor			cool / warm / hot
On top of the cupboard			cool / warm / hot

I work it out. Warmest: _____ °C coolest: _____ °C

Difference: _____ − _____ = _____ °C

When the temperature went up, the humidity went: ☐ ↑ ☐ ↓ ☐ →

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

The difference is explained by _____

_____

Write one sentence: why our room does not have one temperature.

_____

_____

Exit ticket

Write one thing that surprised you today.

_____